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G.L.A. UNIVERSITY, MATHURA

B.TECH. II YEAR III SEM. FIRST MID – TERM EXAMINATION, SEPT. 2011

(BRANCHES - CSE, ECE and ME)

M.M.: 40 **MATHEMATICS - III (AHM – 201)** **Time : 1 Hr. 30 Min**

Instructions :

1. This question paper has **three** parts – A, B and C.
2. Attempt **ALL** parts of this question paper.
3. Marks are indicated against each part.

PART A

Attempt ALL questions of this Part. **(8 x 2 = 16 Marks)**

Q.1 Obtain a cubic equation whose roots are three times of the roots of the equation

$$x^3 - 2x + 10 = 0$$

Q.2 Express the cubic equation $x^3 - 3x^2 + 3 = 0$ in powers of $(x - 1)$.

Q.3 Derive a new cubic equation after removing the second term of the equation

$$x^3 + 6x^2 - 12x + 32 = 0.$$

Q.4 The equation $x^4 - 3x^2 - 42x - 40 = 0$ has one positive, one negative and two imaginary roots. **(True/False)**

Q.5 Fourier series is applicable only for continuous functions . **(True/False)**

Q.6 In expansion of a function f (x) in half range cosine series in $(0, \pi)$, what will be the value of a_0 ?

Q.7 What will be the value of b_n in finding the Fourier series for $f(x) = x^3$ in $(-\pi, \pi)$?

Q.8 Ferrari's method is used to find the roots of a cubic equation. **(True/False)**

P.T.O.

PART B

Attempt any THREE of the following: **(3 x 5 = 15 Marks)**

Q9. Obtain half range cosine series of $F(x) = x$ in $0 \leq x \leq 2$.

Q10. Obtain the half range sine series of $F(x) = \pi x - x^2$ in $(0, \pi)$.

Q11. Solve : $x^3 - 27x + 54 = 0$ by Cardon's method.

Q12. Solve : $x^4 - 5x^2 - 6x - 5 = 0$ by Descartes' method.

PART C

Attempt any ONE of the following questions : **(1 x 9 = 9 Marks)**

Q14. Solve by Cardon's method : $x^3 - 3x^2 + 12x + 16 = 0$.

Q15. If $F(x) = \begin{cases} x, & 0 \leq x \leq \pi \\ 2\pi - x, & \pi \leq x \leq 2\pi \end{cases}$

Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots\ldots\ldots = \frac{\pi^2}{8}$

Q16. Find the Fourier series of $F(x) = x^2$ in $-\pi \leq x \leq \pi$.

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Course: B. Tech. 2nd yr Sem.-III Date: 26-09-11
Subject: Environmental Studies
Paper Code: CEE-201 MM: 40 Time 1.5 h

Q. 1. Attempt all questions (1 x 16)

(i) The environment which has been modified by human activities is called:

(a) Natural environment (b) Anthropogenic environment
(c) Urban environment (d) Modern environment

(ii) The quality of environment has deteriorated due to –

(a) Public awareness (b) NGO

(c) Air, Water and soil pollution (d) None

(iii) Ozone day is celebrated on –

(a) 5th June (b) 14th November

(c) 16th September (d) 28th September

(iv) Which of the following is non biodegradable –

(a) Polythene (b) Vegetable waste

(c) Cotton cloth (d) Wood

(v) Environmental education can be imparted only at –

(a) Primary school stage (b) High school stage
(c) College stage (d) All the levels

(vi) Environmental degradation has been due to-

(a) Poverty of developing countries (b) Poor quality of life (c) Over-consumption of earth`s resources (d) None

(vii) Cloud formation takes place in-

(a) Troposphere (b) Stratosphere
(c) Mesosphere (d) Thermosphere

(viii) O_3 layer present in-

(a) Troposphere (b) Stratosphere
(c) Mesosphere (d) Thermosphere

(ix) The term ecosystem was proposed by-

(a) W.E. Odum (b) A.G. Tansley
(c) H.T. Odum (d) E.P. Odum

(x) Most stable ecosystem is –

(a) Forest (b) Desert (c) Mountain
(d) Ocean

(xi) Energy flow in an ecosystem is always-

(a) Unidirectional (b) Cyclic
(c) Reversible (d) Multi-directional

(xii) In a food chain human are-

(a) Primary consumer (b) Secondary consumer
(c) Top consumer (d) Producer

(xiii) Green plants are also called the-

(a) Autotrophs (b) Producers
(c) Converters (d) All of the above

(xiv) Driving force in an ecosystem is –

(a) Plants (b) Producers
(c) Solar energy (d) Biomass energy

(xv) Secondary consumers are-

(a) Which feeds on herbivores (b) Which feeds on plant
(c) Which feeds on microorganism
(d) Which feeds on dead organism

(xvi) Food webs are –

(a) One kind of food (b) Not related to food chain at all
(c) Interconnected arrangement of food chains
(d) Linear arrangement of food chain

P.T.O.

Q 2. Attempt any four questions **(3 x 4)**

(i) Give a brief account of the importance of environmental studies.

(ii) Differentiate between food chain and food web.

(iii) Describe the structure and function of pond ecosystem.

(iv) Differentiate between ecology and environmental science.

(v) Write short note on biosphere.

(vi) Define essential and non-essential type of environment.

Q 3. Attempt any two questions **(2 x 6)**

(i) With the help of neat sketch discuss the structure of atmosphere along with temperature profile of atmosphere.

(ii) What is scope of environmental science?

(iii) Define an ecosystem. Give an account of the structure and function of ecosystem.

(iv) Write short on-

(a) Atmosphere (b) hydrosphere

(c) Anthropogenic environment

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Data Structure And Algorithms

B.Tech, II[nd] Year, III[rd] Semester, Session- 2011-2012

First Mid-Term Exam

Maximum Marks – 40 Time – 1.5 Hour

Note: - All the Questions are compulsory.

Section A (16 marks)

Q1: Fill in the blanks :- (1*5)

1. An algorithm that calls itself directly or indirectly is known as.....
2. The data structure which allows deletions at both ends of the list but insertion at only one end.....
3. A linked list index is ____ that represents the position of a node in a linked list.
4. Notations that are used to represent the upper bound of complexity is .....
5. Data structure are mainly classified as .....and.....

Q2: Choose the correct answer among following choices :- (1*6)

1. Which of the following is two way list?
   a. grounded header list
   b. circular header list
   c. linked list with header and trailer nodes
   d. none of above
2. The situation when in a linked list START=NULL is
   a. underflow b. Overflow c. Housefull d. Saturated
3. Which of the following data structures are indexed structures?
   a. linear arrays b. linked lists c. both of above d. none of above
4. The memory address of fifth element of an array can be calculated by the formula
   a. LOC(Array[5]=Base(Array)+w(5-lower bound), where w is the number of words per memory cell for the array
   b. LOC(Array[5])=Base(Array[5])+(5-lower bound), where w is the number of words per memory cell for the array
   c. LOC(Array[5])=Base(Array[4])+(5-Upper bound), where w is the number of words per memory cell for the array
   d. None of above
5. The memory address of the first element of an array is called
   a. floor address b. foundation address c. first address d. base address
6. A variable P is called pointer if
   a. P contains the address of an element in DATA.
   b. P points to the address of first element in DATA
   c. P can store only memory addresses
   d. P contain the DATA and the address of DATA

Q3: True or False :- (1*5)

1. A self-referential structure contains a pointer member that points to a structure of the same structure type. For example. the following definition of structure node.

```
struct node {
  int data;
  struct node *nextPtr;
        };
```

2. A stack is a linked-list that can be accessed from either end.
3. Queue nodes are removed only from the front (head) of a queue and are inserted only at the back (tail) of the queue.

4. A tree is an example of a linear data structure.
5. Tower of Hanoi is an example of recursion

Section B (12 Marks)

Q4: Answer any three questions :- **(3*4 = 12)**

1: Consider the postfix expression P given below. Evaluate P using the stack based algorithm.

P: 16, 7, 2, -, /, 2, 1, 5, +, *, +

2: Let a and b denote the positive integers. Suppose a function Q is defined recursively as follows:

$$Q(a, b) = \begin{cases} 0 & \text{if } a < b \\ Q(a-b, b)+1 & \text{if } b \leq a \end{cases}$$

Find the value of Q(2,3) and Q(14,3) .

3:Write an algorithm for matrix multiplication.

4:Write an algorithm to insert an element in stack(using array)

Section C (12 Marks)

Q5: Answer any two questions:- **(2*6 = 12)**

1:Write an algorithm for inserting and deleting an element from the queue.

2: What is a linked list ? Write an algorithm for inserting an element at the end of the link list.

3:Differentiate between doubly linked list and circular linked list. Write an algorithm for traversing a circular linked list.

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Computer Organization

B.Tech, II[nd] Year, III[rd] Semester, Session- 2011-2012

First Mid-Term Exam

Maximum Marks – 40 Time – 1.5 Hour

Note: - All the Questions are compulsory.

Section A (16 marks)

Q1: Fill in the blanks :- **(1*5)**

a) A.....is a combinational circuit that select binary information from one of the many input lines and directs it to a single output lines.

b) A.....is a digital circuit that takes.....input lines and select exactly one of the possible.....output lines.

c) The two's complement representation of –10 is.....

d) The most common error deduction code is the .....

e) The direct communication between processor and the memory of the computer system is implemented with the help of two registers.....and .....

Q2: Choose the correct answer among following choices :- **(1*6)**

1- (a) The task of performing arithmetic and logical operations is called

(i) ALU (ii) editing (iii) storage (iv) output.

2- The ALU and CU jointly known as

(i) RAM (ii) ROM (iii) CPU (iv) none of the above.

3- The circuit used to store one bit of data is known as

(i) Register (ii) Encoder (iii) Decoder (iv) Flip Flop

4- _____ register keeps track of the instructions stored in program stored in memory.
(i) AR (Address Register) (ii) XR (Index Register)(iii) PC (Program Counter) (iv) AC (Accumulator).

5- A binary digit is called a
(i) Bit (ii) Byte (iii) Number (iv)Character.

6- The gray code equivalent of (1011)2 is
(i) 1101. (ii) 1010.(iii) 1110. (iv) 1111

Q3: True or False :- **(1*5)**

a) Registers contain data and instructions needed by the CPU.

b) The fetch-decode-execute cycle refers to the process by which data is read from the hard drive and stored in memory.

c) Terminal is a point at which data enters or leaves the computer

d) A bootstrap is an error correction technique

e) Half adder is logic CKT that add three Digit at a time

PTO

Section B (12 Marks)

Q4: Answer any three questions :- **(3*4 = 12)**

1. The content of a 4-bit register is initially 1101. The register is shifted six times to the right with the serial input being 101101. What is the content of the register after each shift.
2. Draw the logic diagram of a 2 to 4 line decoder with only Nor gated.Include an enable input.
3. Elaborate on different types of registers in a register organization

Section C (12 Marks)

Q5: Answer any two questions:- **(2*6 = 12)**

1. Draw and Explain the instruction cycle with interrupts.
2. Simplify the Boolean fuction F together with the don't care conditions d in (1) sum-of-product form And (2) Product-of-sums form.
   $F(w,x,y,z)=\sum(0,1,2,3,7,8,10)$
   $d(w,x,y,z)=\sum(5,6,11,15)$
3. Construct a 16 to 1 line multiplexer with two 8-to-1 multiplexer and one 2-to-1 line multiplexer. Use block diagram for the three multiplexers.

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DISCRETE STRUCTURES (CSE - 213)

B.Tech, CS, II[nd] Year, III[rd] Semester, Session- 2011-2012

First Mid-Term Exam

Maximum Marks – 40 | Time – 1.5 Hour

Note: - All the Questions are compulsory.

Section A (16 marks)

Q 1. Choose the correct alternative: **(9*1=9)**

A) Number of elements in a singleton set is
- a) 0
- b) 1
- c) Any natural number
- d) None of these

B) Which one of the following pairs of sets comprises of disjoint sets
- a) {1, 2, 3} and {2, 4, 7 ,9}
- b) {0, 1} and {1, 2}
- c) {0} and {1}
- d) None of these

C) A – B is equal to
- a) A U B
- b) A ∩ B
- c) A ∩ B'
- d) None of these

D) If A = {3, 5} then A X A is equal to
- a) {(3, 3),(3, 5),(5, 3),(5, 5)}
- b) {(3, 3),(5, 5)}
- c) { (3, 5), (3, 3), (5, 5)}
- d) None of these

E) R is transitive if
- a) aRb and bRc =>aRc
- b) cRa
- c) aRc
- d) None of these

F) Which of the following is a tautology?
- a) (p ^ (p)')'
- b) p ^ p
- c) p' ^ p'
- d) p v p

G) Negation of the statement "He is neither intelligent nor a player" is
- a) He is intelligent or a player.
- b) He is intelligent and a player.
- c) He is intelligent but not a player.
- d) He is not intelligent but a player.

If p denotes " he is intelligent" and q denotes "He is a player", then find the correct formula to represent the statement given below:

H) Either he is not intelligent or he is not a player.
- a) p' v (p ^ q')
- b) p' v q
- c) p' ^ q'
- d) p' v q'

I) He is intelligent if and only if he is a player.
- a) p ↔ q
- b) p → q
- c) q → p
- d) p' → q'

Q 2. Fill in the blanks: **(6*1=6)**

A) If two sets P and Q have 3 elements in common and n(P) = 7, n(Q) = 5, then n(P X Q) is equal to.....

B) A U A' is equal to .....

C) A ∩ Ø is equal to .....

D) The number of rows in a truth table of a compound statement having n distinct primary or atomic statements is .....

E) p → q is false when p is ..... and q is .....

F) p v q is false when p is ..... and q is .....

G) If p is true and q is false then $p \leftrightarrow q$ is .....

Section B (12 Marks)

Q 3. Attempt any Three:

A) Detemine the domain and range of the relation: (4)
$\{(x, y) : x^2 + y^2 = 25, x \text{ and } y \text{ are integers}\}$

B) Prove any two of the following: (2*2=4)

a) If $A \subseteq B$ then $A \cup B = B$

b) $X - (Y \cap Z) = (X - Y) \cup (X - Z)$

c) $\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

C) Given that the ordered pairs (a, 7) and (-4, b) belong to the relation $\{(x, y) : 2y-3x = 8\}$. Find the values of a and b. (4)

D) Attempt any one: (4)

a) Obtain the principle DNF of $\neg(p \rightarrow (q \wedge r))$.

b) Obtain the principle CNF of $(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$.

SECTION – C (12 MARKS)

Q 4. Attempt any two:

A) Suppose that 100 of the mathematics students at a college take at least one of the languages French, German and Russian. Also suppose: (2+4 = 6)

60 study French
40 study German
37 study Russian
15 study French and German
20 study French and Russian
10 study German and Russian

Draw a Venn diagram and fill in the correct number of students in each region. Also determine the number of students who study

a) All the three languages.
b) French or German but not Russian.
c) Only Russian.
d) French and Russian but not German.

B) Which of the following relations are: (1.5*4=6)

a) Reflexive
b) Symmetric
c) Anti-symmetric
d) Transitive

defined on A = {1, 2, 3}

R1 = {(1, 1), (1, 2), (1, 3), (3, 3)}

R2 = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)}

R3 = {(1, 1), (1, 2), (2, 2), (2, 3)}

A X A (Universal Set)

Ø (Null set)

C) **(i)** Check the validity of the following argument: (3+3 = 6)

" If Roli has completed MCA or MBA, then she is assured of a good job. If Roli is assured of a good job, she is happy. Roli is not happy. So, Roli has not completed MBA"

(ii) Show that $\neg r$ is a valid conclusion from the premises $p \rightarrow \neg q$, $r \rightarrow p$, q using the method of truth table.

B.Tech I I Year, III Semester
First Mid Term Test-2011
Object Oriented Programming
[ITE 211]

Time: 1Hour 30 minutes **Total Marks: 40**

Note- Attempt all questions.

Q1. Choose the most appropriate option. **[16x1=16]**

1. Object Oriented Programming follows:-
   (a) Top-down approach (b) Bottom -up Approach
   (c)Sequential Approach (d) None of these
2. Object based Programming does not have
   (a)Polymorphism (b)Inheritance
   (c)Dynamic Binding (d) Both (b) and (c)
3. The process of making a Function to exhibit different behaviours in different instances is comes under the concept of
   (a)Inheritence (b) Data Abstraction
   (c)Polymorphism (d)Data Hiding
4. C++ was developed by
   (a)Dennis Ritchie (b)James Gosling
   (c)E Balagurusamy (d)Bjarne Stroustrup
5. In the Statement cout<<"Hello"; what is the name of << operator
   (a)Left Shift operator (b)Right Shift operator
   (c)Insertion Operator (d)Extraction operator
6. Default return type of a function is
   (a) int (b) float
   (c) char (d) void
7. The program execution starts from
   a) The function which is first defined
   b) The function which is last defined
   c) main() function
   d) the function other than main()
8. The function main ( ) is:
   a) a built-in function
   b) a user defined function

c) Both

d) None of these

9. class is a

(a)Derived Datatype (b)Primary Datatype

(c)User-defined Datatype (d) None

10. By default ,the members of the class are

(a) public (b) private

(c)protected (d) None of these

(B) Fill in the blanks:-

11. The operator :: is known as _____ operator
12. While(i=100) {.....} results in an _____loop.
13. The wrapping of data and function into single unit is called _____.

(C) True or False:-

14. A function can be called from more than one place within a program.
15. An Array should be declared before use.
16. Member function can not access the private data of the same class.

Q2.Attempt any three of the following [4*3=12]

(a). What are the differences between Procedure-oriented Programming and OOP ?

(b). Write a c++program to find the exact ratio of males and females.The number of males and females are provided by the user and is of integer type.

(c). Explain Function overloading with example.

(d). What are the advantages of using new operator as compared to function malloc()?.

Q3.Attempt any two of the following [6*2=12]

(a). Explain the difference between call by value and call by reference with examples.

(b) Define a class to represent a bank account .Include the following members:

Data Members:1)Name of the depositor
2)Account Number
3)Balance Amount in the account

Member Functions:1)To assign initial values
2)To deposit an amount
3)To withdraw an amount after checking the balance
4)To display name and balance

(c) Write a short note on:-

(i) Encapsulation

(ii) Data Hiding.

(iii)Inheritance